

CA_02 REPORT

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dpu.py

Declares the SimObject **DataProcessingUnit** and connects the C++ class to Python. It exposes the object to gem5's configuration system via:

- `type = 'DataProcessingUnit'`
- `cxx_class = "gem5::DataProcessingUnit"`
- `cxx_header = "learning_gem5/part2/dpu.hh"`

This lets us instantiate and configure the object from Python run scripts.

dpu.hh

C++ header declaring the **gem5::DataProcessingUnit** class, which inherits from SimObject. The header is wrapped in include guards and includes `params/DataProcessingUnit.hh`. It declares the task methods (string reversal, factorial, sorting) and EventFunctionWrapper members for scheduling them.

dpu.cc

C++ implementation of the functions declared in `dpu.hh`. It defines the logic for reversing a string, computing a factorial, and sorting an array, schedules events in `startup()`, and uses `DPRINTF` with debug flags (**REVSTRING**, **FACTORIALRESULT**, **SORTRESULT**) so intermediate results can be traced. It ends by calling `exitSimLoop` after finishing the tasks.

SConscript

Builds and registers the SimObject and sources:

- `SimObject('dpu.py', sim_objects=['DataProcessingUnit'])`
- `Source('dpu.cc')`

It also declares the debug flags used by the implementation, ensuring the object builds correctly and that runtime logging can be enabled.

Run_dpu.py

Python configuration script that creates a System, sets clock/voltage domains, instantiates DataProcessingUnit, and runs the simulation (`m5.instantiate()`, `m5.simulate()`). Running this verifies the implemented operations and prints the exit cause.

Command to run

`build/X86/gem5.opt --debug-flags=REVSTRING,FACTORIALRESULT,SORTRESULT \`
`configs/learning_gem5/part2/run_dpu.py`
(Add `-d m5out` if you want a specific output directory.)

```
gem5 Simulator System. https://www.gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 25.0.0.1
gem5 compiled Sep 15 2025 03:13:34
gem5 started Sep 15 2025 03:18:39
gem5 executing on Sarfaraz, pid 19427
command line: build/X86/gem5.opt --debug-flags=REVSTRING,FACTORIALRESULT,SORTRESULT configs/learning_gem5/part2/run_dataproc.py

Global frequency set at 1000000000000 ticks per second
warn: No dot file generated. Please install pydot to generate the dot file and pdf.
Enter a string for reversal: nischal
Enter a number for factorial: 7
Enter number of elements for sorting: 5
Enter 5 integers: 10 96 5 23 7
0: system.dpu: User provided factorial number: 7
0: system.dpu: User provided array: 10 96 5 23 7
Beginning simulation with DataProcessingUnit...
Enter tick for String Reversal: 200
Enter tick for Factorial: 1500
Enter tick for Sorting: 2000
200: system.dpu: Original: nischal, Reversed: lahcsin
1500: system.dpu: Factorial of 7 is 5040
2000: system.dpu: Sorted array: 5 7 10 23 96
Exiting @ tick 2000 because Completed DataProcessingUnit tasks
```